

VIDIN, Bulgaria

WMO: 155020

Lat: 43.983N

Lon: 22.850E

Elev: 31

StdP: 100.95

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-9.5	-15.0	1.0	-12.8	-11.4	1.4	-9.0	9.1	3.5	7.1	3.9	0.8	320	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	34.4	22.9	32.8	22.4	31.2	21.9	24.2	31.8	23.3	30.8	22.5	29.8	2.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.5	28.5	20.9	15.6	27.3	20.1	14.8	26.3	72.9	31.9	69.4	30.9	66.4	29.9	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.3	5.4	4.7		-17.0	36.9	4.3	2.1	-20.2	38.5	-22.7	39.7	-25.1	40.9	-28.3	42.4
				-17.3	26.3	4.3	1.1	-20.4	27.1	-23.0	27.7	-25.4	28.3	-28.5	29.2
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-0.2	2.2	7.4	12.6	17.5	21.4	23.5	23.2	18.3	12.0	6.3	1.0		
	DBStd	9.22	5.17	5.48	4.56	3.41	3.10	2.98	2.51	2.67	3.32	3.98	4.00	4.54		
	HDD10.0	1086	317	222	104	14	0	0	0	0	26	123	280			
	HDD18.3	2743	574	452	340	174	53	8	1	2	42	200	361	537		
	CDD10.0	1871	1	3	23	92	232	342	420	409	248	87	13	1		
	CDD18.3	486	0	0	0	2	27	99	162	153	40	3	0	0		
	CDH23.3	5384	0	0	4	44	338	1098	1789	1646	426	37	1	0		
	CDH26.7	2174	0	0	0	7	71	421	801	743	127	5	0	0		
(14) Wind	WSAvg	1.6	1.5	1.9	2.1	1.8	1.6	1.6	1.5	1.3	1.4	1.3	1.4	1.5		
(15) Precipitation	PrecAvg	562	39	36	39	48	56	54	53	42	49	50	44	51		
	PrecMax	986	77	85	88	129	115	130	124	171	143	150	103	132		
	PrecMin	276	7	1	4	0	23	9	5	1	3	4	0	0		
	PrecStd	148	21	26	25	25	25	31	35	39	36	37	29	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.2	18.7	23.2	27.7	30.6	35.1	37.3	36.7	32.6	27.2	20.8	15.5		
		MCWB	9.7	11.8	13.7	17.3	21.0	22.9	22.7	23.4	20.9	18.5	14.5	10.3		
	2%	DB	11.6	15.6	20.4	24.2	28.6	32.8	34.4	34.3	29.9	23.8	16.7	11.1		
		MCWB	7.6	10.1	12.4	15.9	19.8	22.7	22.9	22.9	20.0	17.2	12.5	7.7		
	5%	DB	8.8	12.6	17.9	21.8	27.0	30.9	32.7	32.7	27.9	21.3	14.2	8.6		
		MCWB	6.0	8.4	11.2	14.7	19.0	22.2	22.5	22.3	19.4	15.9	11.5	6.2		
	10%	DB	6.3	9.8	15.3	19.9	25.1	29.1	31.0	31.0	25.8	18.9	12.2	6.8		
		MCWB	4.5	6.6	9.9	13.8	18.3	21.4	22.0	21.8	18.6	14.6	10.5	5.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	12.2	14.6	18.3	22.0	24.9	25.1	25.5	22.2	19.6	15.4	10.6		
		MCDB	14.7	18.1	21.7	25.3	29.0	32.2	32.9	32.9	29.9	25.5	19.4	14.8		
	2%	WB	7.8	10.2	12.9	16.7	20.6	23.7	24.0	24.2	20.8	17.8	13.3	8.2		
		MCDB	11.2	14.9	19.2	23.1	27.1	30.9	31.9	31.7	27.8	22.6	15.8	10.6		
	5%	WB	6.2	8.6	11.7	15.4	19.6	22.7	23.2	23.1	19.8	16.3	12.0	6.6		
		MCDB	8.3	12.0	17.2	20.8	25.7	29.6	30.9	30.6	26.3	20.6	13.5	8.1		
	10%	WB	4.7	7.0	10.3	14.2	18.6	21.7	22.4	22.2	19.0	15.0	10.7	5.3		
		MCDB	6.3	9.5	14.8	18.9	24.2	28.1	29.7	29.5	24.8	18.4	12.1	6.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	8.5	10.9	11.8	12.1	12.7	13.4	13.8	12.2	10.2	7.2	6.9		
		MCDBR	11.6	14.0	16.1	15.9	15.8	16.2	16.8	17.2	16.4	14.9	12.1	10.6		
	5% WB	MCWBR	7.9	9.3	9.4	8.6	7.9	7.3	6.8	7.1	7.6	8.7	8.0	7.5		
		MCWBR	10.6	13.1	14.8	14.3	14.4	14.9	14.9	15.1	14.4	13.7	9.8	9.7		
(40) Clear-Sky Solar Irradiance	taub		0.347	0.375	0.414	0.476	0.456	0.450	0.461	0.479	0.448	0.405	0.373	0.346		
	taud		2.285	2.199	2.165	2.016	2.138	2.198	2.169	2.115	2.165	2.276	2.313	2.310		
	Ebn at Noon		753	794	813	790	821	826	811	779	770	754	716	714		
	Edh at Noon		86	111	131	165	150	142	145	148	130	102	84	77		
(44) All-Sky Solar Radiation	RadAvg		1.41	2.22	3.38	4.56	5.56	6.27	6.33	5.64	4.03	2.64	1.53	1.19		
	RadStd		0.20	0.32	0.44	0.52	0.49	0.53	0.43	0.49	0.35	0.37	0.21	0.17		

Historical Trends

		Heating		Cooling		Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.59	N/A	N/A	N/A	+0.38	N/A	-149	+93	N/A

Nomenclature: See separate page